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Batch: A-2
Division: A

Subject: Programming with Java

ASSIGNMENT No : 6

TITLE : Write a Java program to demonstrate the use of an inner class, an anonymous class.

Problem Statement

Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Theory

An inner class is a class defined inside another class and has access to all members of the enclosing class, including private members. Inner classes help improve code organization and logical grouping of related classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class definitions. These concepts are widely used in Java GUI programming and event handling.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp6/Main.java
Inner and Outer classes :
Inner Class Method
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ 
```

Exercise

Q1. Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp6/Exercise/Ex1.java
Car Name : Toyota
Car Model : Etios
Car started!
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp6/Exercise/Ex2.java
Order ID      : 101
Customer Name : Devansh
Food Item     : Pizza
Delivery Status : Out for Delivery
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

This assignment helped in understanding the concepts of inner classes and anonymous classes in Java. It demonstrated how an inner class can access the members of its enclosing class to improve code organization and encapsulation, while an anonymous class can provide a one-time implementation of an interface or abstract class without creating a separate class. The implementation improved my understanding of object-oriented programming concepts, event handling, and code reusability. These concepts are widely used in Java GUI development, event-driven programming, multithreading, and real-world applications where concise and modular code is required.